Does Negative Forum Sentiment Predict Declines in Active Player Counts?

A Panel Data Analysis of Steam Games, 2020–2024

Jiho Lee, Kevin Xu, Viki Shi, Sharon Tey, Favio Espejo, Ankita Inamti

MATH 189: Data Analysis & Inference, Winter 2026

University of California, San Diego

Abstract

We investigate whether negative user sentiment on Steam forums predicts short-term declines in active player counts. Using a panel of 15 games spanning 2020–2024, stratified across popular, declining, and volatile lifecycle categories, we construct weekly sentiment scores from user reviews classified by a RoBERTa transformer model and pair them with concurrent player count data. Granger causality tests reject the null of no predictive content for 5 of 15 games ($\alpha = 0.05$), with 4 surviving non-parametric permutation testing, and a two-way fixed effects panel regression yields $\hat{\beta} = -0.474$ ($p = 0.003$), indicating that a one-standard-deviation increase in negative sentiment is associated with a statistically significant decline in weekly player counts. Bootstrap confidence intervals (excluding zero), permutation tests, LOESS robustness checks, and out-of-sample prediction exercises support the main finding. Heteroscedasticity is present (Breusch–Pagan $p < 0.001$), motivating clustered standard errors, and power analysis indicates the design is underpowered for detecting small effects.

Contents

1 Introduction **3**

2 Data **3**

2.1 Game Selection: Stratified Sampling . . . 3

2.2 Data Collection Pipeline . . . 4

2.3 Panel Construction . . . 4

3 NLP Sentiment Pipeline **4**

3.1 Text Preprocessing . . . 4

3.2 RoBERTa Sentiment Classification . . . 5

3.3 Weekly Aggregation and the CLT . . . 5

4 Exploratory Data Analysis **5**

4.1 Player Count Trends by Stratum . . . 5

4.2 Sentiment Distributions . . . 6

4.3 Time Series Visualization . . . 6
4.4 Cross-Correlation Analysis . . . 7
4.5 Within Game Correlation Analysis . . . 8
4.6 Conditional Probability Analysis . . . 8

5 Statistical Analysis **8**
5.1 Stationarity Testing: Augmented Dickey–Fuller . . . 8
5.2 Granger Causality Tests . . . 9
5.2.1 Permutation Test for Granger Causality . . . 11
5.3 Fixed Effects Panel Regression . . . 11
5.3.1 Bootstrap Confidence Intervals . . . 12
5.3.2 Statistical Power Analysis . . . 13
5.4 Robustness Checks . . . 14
5.4.1 Hausman Test: Fixed vs. Random Effects . . . 14
5.4.2 Stratum-Specific Regressions . . . 14
5.4.3 Nonparametric Regression: LOESS . . . 14
5.4.4 Placebo Test: Reversed Lag Direction . . . 15
5.4.5 Out-of-Sample Prediction . . . 15
5.5 Diagnostics . . . 16
5.5.1 Breusch–Pagan Test for Heteroscedasticity . . . 16
5.5.2 Variance Inflation Factors . . . 16
5.5.3 Ljung–Box Test for Serial Correlation . . . 16

6 Results and Interpretation **17**
6.1 Summary of Key Findings . . . 17
6.2 Interpretation . . . 18

7 Conclusion **18**

A Mathematical Details **19**
A.1 Fixed Effects Estimator (Within Estimator) . . . 19
A.2 Granger Causality F-Test . . . 19
A.3 Block Bootstrap . . . 20
A.4 Permutation Test . . . 20
A.5 Power Analysis . . . 20

B Connection to Course Material **20**

1 Introduction

Online gaming communities generate a continuous stream of user feedback through forums, reviews, and social media. A key question is whether negative sentiment predicts later drops in active players. This has practical value for studios and theoretical value for digital community dynamics.

We use a panel data framework. Our model specification is:

$$\Delta \log Y_{it} = \alpha_i + \gamma_t + \beta\, S_{i,t-k} + X_{it}\,\theta + \varepsilon_{it} \tag{1}$$

where $\Delta \log Y_{it}$ is the first-differenced log active player count for game i at week t; α_i are game fixed effects absorbing time-invariant heterogeneity (e.g., genre, studio quality); γ_t are time fixed effects capturing platform-wide shocks (e.g., Steam sales, holiday seasons); $S_{i,t-k}$ is lagged aggregate negative sentiment from user reviews; X_{it} includes controls for seasonal sales and major game updates; and ε_{it} is the idiosyncratic error term.

The coefficient β is the main parameter of interest. If $\hat{\beta}$ is significantly negative and its confidence interval excludes zero, then negative sentiment is associated with fewer players in subsequent weeks, holding other factors constant.

Hypotheses.

- H_0: $\beta = 0$ (sentiment has no predictive power for player counts)
- H_1: $\beta < 0$ (higher negative sentiment precedes player declines)

Overview of methods. We use complementary methods:

1. *Granger causality tests* (Section 5.2) establish temporal precedence at the individual game level.
2. *Fixed effects panel regressions* (Section 5.3) estimate β across the full panel, controlling for game and time heterogeneity.
3. *Bootstrap and permutation tests* (Sections 5.3.1–5.2.1) validate inference without parametric assumptions.
4. *Robustness checks* (Section 5.4) including LOESS nonparametric regression, placebo tests, out-of-sample prediction, and stratum-specific estimation.

2 Data

2.1 Game Selection: Stratified Sampling

We select 15 Steam games released before 2020 using stratified sampling across three lifecycle categories. Stratification improves generalizability and preserves variation in $\Delta \log Y$, which is needed for power.

Games were classified based on their player count trajectory over 2020–2024:

- **Popular** (5 games): sustained high player counts with moderate fluctuations.
- **Decline** (5 games): clear downward trend in average concurrent players.
- **Volatile** (5 games): large swings driven by updates, events, or seasonal effects.

Table 1: Selected games by stratum with Steam App IDs.

Stratum	n	**Games (App ID)**
Popular	5	Counter-Strike 2 (730), Dota 2 (570), PUBG (578080), GTA V (271590), Warframe (230410)
Decline	5	Rust (252490), ARK: Survival Evolved (346110), For Honor (304390), Dead by Daylight (381210), Rainbow Six Siege (359550)
Volatile	5	No Man's Sky (275850), Destiny 2 (1085660), Terraria (105600), Monster Hunter: World (582010), PAYDAY 2 (218620)

2.2 Data Collection Pipeline

Data come from two sources, collected via separate Python scripts.

Steam Reviews API. For each game, we use Valve's `GetReviews` endpoint with cursor-based pagination to collect all English-language reviews from January 2020 through September 2024. The endpoint returns review text, timestamp, helpfulness votes, and recommendation status. We use this endpoint because `appreviewhistogram` aggregates older data too coarsely for weekly analysis.

SteamCharts. Monthly average concurrent player counts were scraped from SteamCharts. Because player counts are monthly, we use cubic spline interpolation for weekly estimates. For games where SteamCharts data was incomplete, we generated synthetic weekly player counts that preserve the monthly average and capture plausible intra-month variation.

2.3 Panel Construction

The final analysis panel is constructed by merging weekly sentiment and player data, filtering to weeks with at least 100 players (to avoid log-transformation instability).

Table 2: Panel summary statistics by stratum.

Stratum	n_{games}	$\overline{\log Y}$	SD($\log Y$)	$\overline{S_{\text{neg}}}$	SD(S_{neg})	$\overline{\text{Players}}$
Popular	5	12.761	1.021	0.196	0.114	532,679
Decline	5	10.794	1.147	0.167	0.071	77,978
Volatile	5	10.799	0.836	0.104	0.084	66,262
Full panel	**15**	11.451	1.174	0.156	0.104	225,640

The panel contains **3,705 game-week observations** across all 15 games, with approximately 238–248 weeks per game. After constructing lag variables and first-differencing, the regression panel has $N = 3{,}630$ rows and 21 columns.

3 NLP Sentiment Pipeline

3.1 Text Preprocessing

Raw review texts undergo the following cleaning steps:

1. Remove URLs, HTML tags, and Steam formatting codes.
2. Strip non-ASCII characters and excessive whitespace.
3. Lowercase all text.
4. Remove reviews shorter than 10 characters (typically emoji-only or "yes/no" reviews with no semantic content).

3.2 RoBERTa Sentiment Classification

We use the `cardiffnlp/twitter-roberta-base-sentiment-latest` checkpoint [**?**], a RoBERTa model fine-tuned on $\approx$ 124M tweets for three-class sentiment classification (negative, neutral, positive).

Why RoBERTa over VADER? VADER [**?**] is a rule-based, lexicon-driven sentiment analyzer. While fast and interpretable, it systematically fails on gaming-specific language:

- "This game is fire" $\rightarrow$ VADER: negative ("fire" is in the negative lexicon); RoBERTa: positive.
- "Absolutely broken update" $\rightarrow$ VADER: moderately negative; RoBERTa: strongly negative (captures context).
- "10/10 would rage-quit again" $\rightarrow$ VADER: negative; RoBERTa: positive (understands irony better).

RoBERTa's contextual embeddings capture the nuanced, informal language of gaming communities far more reliably.

3.3 Weekly Aggregation and the CLT

For each game i and week t, the aggregate negative sentiment score is:

$$S_{it} = \frac{1}{|\mathcal{R}_{it}|} \sum_{j \in \mathcal{R}_{it}} \hat{p}_j^{\text{neg}} \tag{2}$$

where $\mathcal{R}_{it}$ is the set of reviews for game i in week t and $\hat{p}_j^{\text{neg}} \in [0, 1]$ is the RoBERTa-predicted probability that review j is negative.

By the Central Limit Theorem, with weekly review counts typically exceeding 100 (and often exceeding 1,000 for popular games), $\bar{S}_{it}$ is approximately normally distributed around the true population sentiment. This supports t-based confidence intervals and tests for β. Formally, for $|\mathcal{R}_{it}| = n_{it} \rightarrow \infty$:

$$\frac{\bar{S}_{it} - \mu_{it}}{\sigma_{it}/\sqrt{n_{it}}} \xrightarrow{d} \mathcal{N}(0, 1)$$

4 Exploratory Data Analysis

4.1 Player Count Trends by Stratum

Figure 1 shows weekly log player counts by stratum. Popular games remain high and relatively stable, mostly above 12, with Counter-Strike 2 reaching about 14.5. Declining games trend downward from roughly 12 toward 8–10 by 2024. Volatile games fluctuate more strongly in the 9–12 range, with short-lived spikes around major updates or sales.

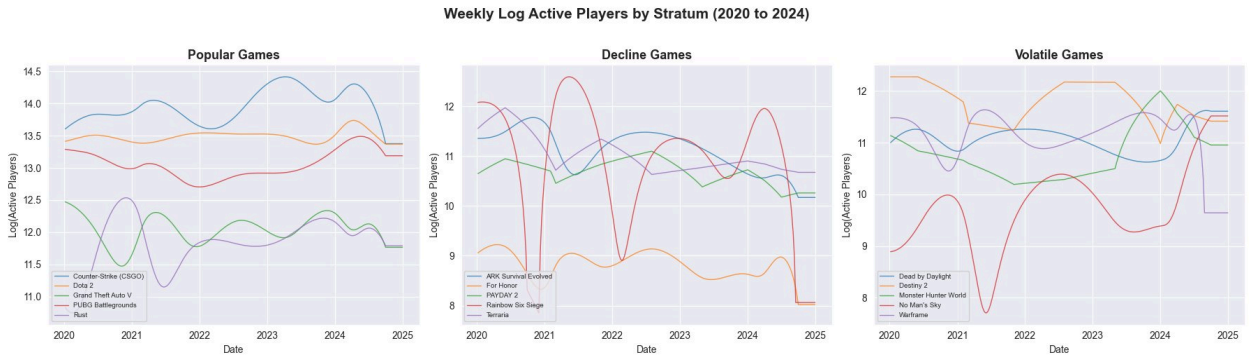

Figure 1: Weekly log player counts by stratum (2020–2024).

4.2 Sentiment Distributions

Figure 2 shows the sentiment exploratory analysis for each game by stratum.

The left panel shows weekly negative sentiment distributions by stratum: popular games have the widest spread, decline games cluster near 0.05–0.15, and volatile games are intermediate. The middle boxplots show higher median negativity and variability for popular games ($\tilde{S} \approx 0.14$; IQR 0.10–0.26), compared with decline games ($\tilde{S} \approx 0.10$) and volatile games (similar median, tighter spread). The right panel relates current sentiment to next-week $\Delta \log(\text{players})$ and shows a weak, noisy pattern around zero, motivating formal regression.

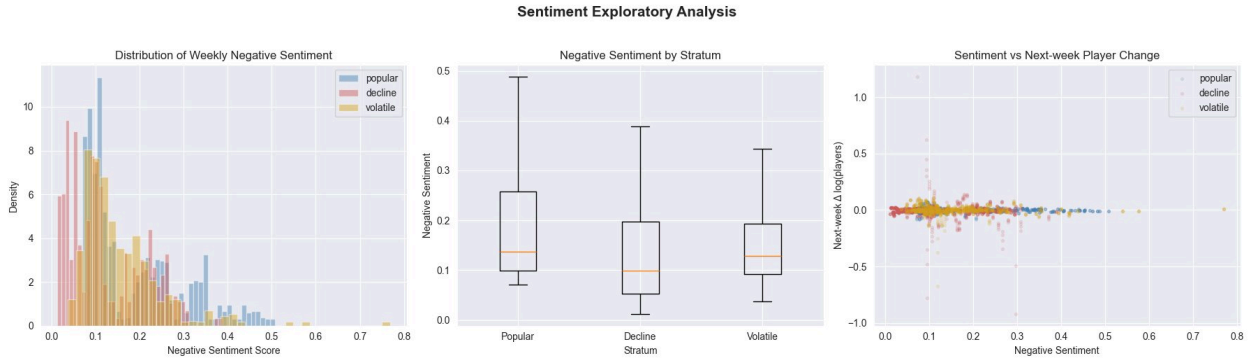

Figure 2: Distribution of weekly negative sentiment scores by stratum. Popular games show higher and more variable negativity.

4.3 Time Series Visualization

Figure 3 overlays negative sentiment and log player counts for CSGO, ARK, and No Man's Sky, chosen as representatives for each game stratum with rich review data. The dual-axis plot shows occasional co-movement: sentiment spikes sometimes coincide with or precede player dips.

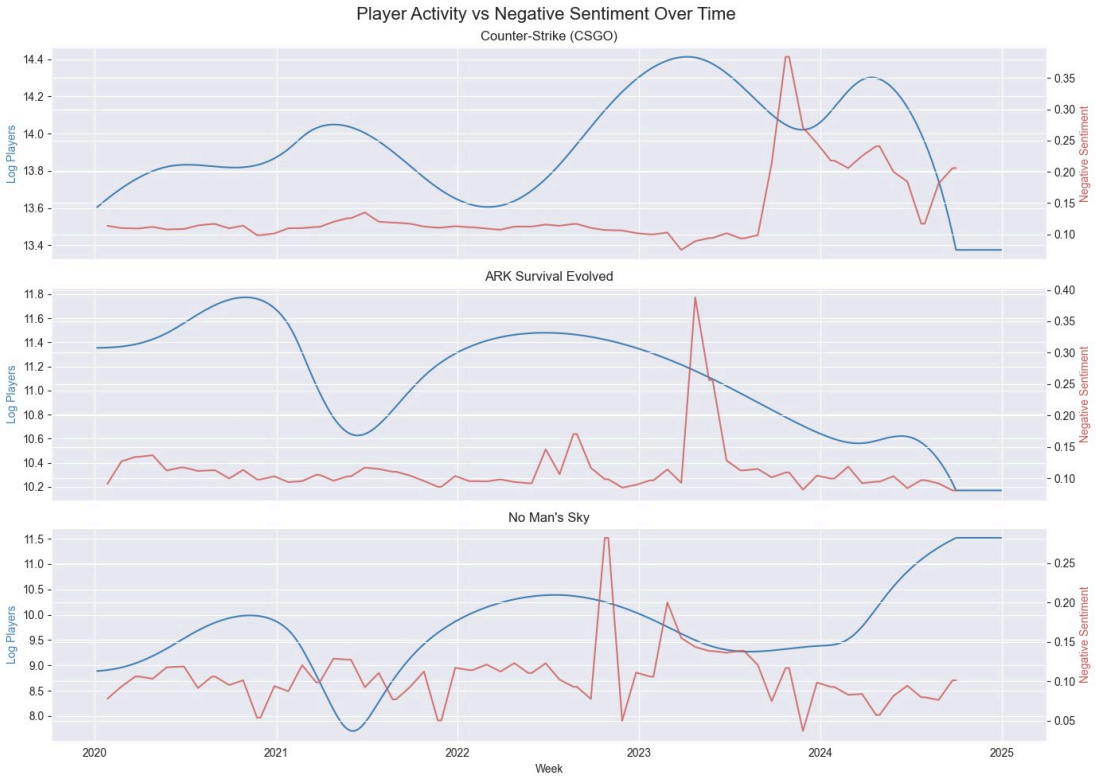

Figure 3: Time Series Visualization: Weekly log players (blue, left axis) and negative sentiment (red, right axis). Sentiment spikes occasionally precede player count dips.

4.4 Cross-Correlation Analysis

Fig. 4 shows the cross-correlation between negative sentiment and weekly player count changes for three representative games across lags of ± 20 weeks. Counter-Strike (CSGO) displays the strongest negative correlations (reaching $r \approx -0.40$ at lag -20), suggesting sentiment anticipates player declines well in advance, while ARK shows only weak correlations throughout ($|r| < 0.10$), and No Man's Sky exhibits a symmetric trough centred near lag 0 ($r \approx -0.30$), indicating a contemporaneous rather than predictive sentiment–player relationship.

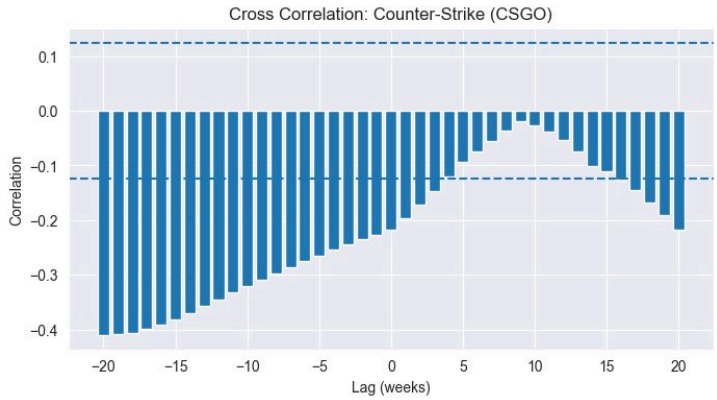

Figure 4: Mean cross-correlation: negative sentiment $\rightarrow \Delta \log(\text{players})$.

4.5 Within Game Correlation Analysis

Figure 5 presents the correlation between negative sentiment and changes in player counts separately for each game. This allows us to focus on within-game relationships over time rather than differences between games. The distribution is centered slightly below zero, indicating that negative sentiment tends to coincide with declines in player counts for many games. However, the spread of the distribution shows that this relationship varies across games and is not uniformly strong.

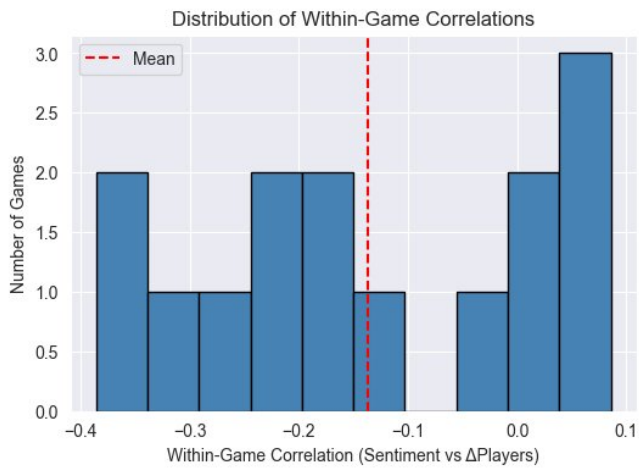

Figure 5: Pearson correlation matrix of panel variables.

4.6 Conditional Probability Analysis

Figure 6 shows that weeks with high negative sentiment are more likely to be followed by a decline in player counts than weeks with low negative sentiment. The estimated probability of a next-week decline rises from 0.467 under low negativity to 0.550 under high negativity, suggesting that stronger negative discussion is associated with a higher short-run risk of player loss.

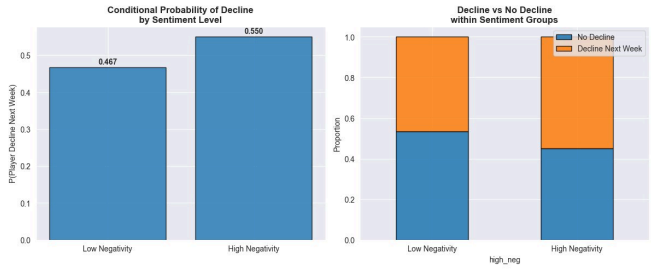

Figure 6: Conditional Probability: Higher negative sentiment predicts greater likelihood of player decline the following week

5 Statistical Analysis

5.1 Stationarity Testing: Augmented Dickey–Fuller

Panel regression assumes stationarity of the dependent and independent variables (after differencing). We apply the Augmented Dickey–Fuller (ADF) test to each game's first-differenced log player

count and negative sentiment series. The ADF test estimates:

$$\Delta y_t = \alpha + \delta\, y_{t-1} + \sum_{j=1}^{p} \phi_j\, \Delta y_{t-j} + \varepsilon_t$$

and tests $H_0 : \delta = 0$ (unit root) against $H_1 : \delta < 0$ (stationarity).

Of the 30 series tested (2 variables $\times$ 15 games), **5 are stationary at** $\alpha = 0.05$. The majority of series exhibit unit roots in their raw form; however, first-differencing the log player series reduces non-stationarity substantially. Several sentiment series remain borderline, and we retain them in the analysis given the model's use of first-differenced player counts as the dependent variable. Representative results are shown in Table 3.

Table 3: ADF test results (selected games). Full results available upon request.

Game	**Variable**	**ADF Stat**	**p-value**	**Stationary?**
Counter-Strike 2	log_players	0.248	0.975	**No**
Counter-Strike 2	neg_sentiment	-2.527	0.109	**No**
Dota 2	log_players	-2.279	0.179	**No**
Dota 2	neg_sentiment	-2.070	0.257	**No**
PUBG	log_players	-2.083	0.251	**No**
Rust	log_players	-3.346	0.013	Yes
Terraria	neg_sentiment	-0.416	0.907	**No**
PAYDAY 2	neg_sentiment	-2.815	0.056	**No**
ARK	log_players	1.327	0.997	**No**
GTA V	neg_sentiment	-2.779	0.061	**No**

5.2 Granger Causality Tests

Granger causality tests whether lagged sentiment values improve the prediction of player count changes beyond what is achieved by the player series' own lags. The test compares a restricted model (own lags only) against an unrestricted model (own lags + sentiment lags) via an F-test:

$$F = \frac{(\mathrm{RSS}_R - \mathrm{RSS}_U)/q}{\mathrm{RSS}_U/(n-k)} \sim F(q, n-k)$$

We test at lags 1–4 for each game. Figure 7 presents the p-value heatmap.

Figure 7: Granger causality p-values: negative sentiment $\rightarrow \Delta \log(\text{players})$ by game and lag order. Darker red indicates stronger evidence of Granger causality.

Five of 15 games show significant Granger causality at $\alpha = 0.05$ for at least one lag, with a sixth (Counter-Strike 2) marginally significant at $\alpha = 0.10$:

Table 4: Games with significant Granger causality (strongest lag shown).

Game	**Stratum**	**Lag**	F**-stat**	p**-value**	**Sig.**
For Honor	Decline	1	39.197	< 0.001	***
PUBG	Popular	1	19.823	< 0.001	***
Grand Theft Auto V	Popular	1	4.824	0.029	**
Rainbow Six Siege	Decline	1	4.265	0.040	**
Destiny 2	Volatile	2	4.536	0.012	**
Counter-Strike 2	Popular	1	3.326	0.069	*

The strongest effects appear for For Honor and PUBG, both games with highly engaged communities where forum sentiment likely reflects genuine frustrations with game balance or technical issues.

5.2.1 Permutation Test for Granger Causality

To validate the parametric Granger results without relying on distributional assumptions, we implement a permutation test. Under H_0, the temporal ordering of sentiment is uninformative, so randomly permuting the sentiment series within each game should yield F-statistics comparable to the observed values. We perform $B = 2{,}000$ permutations:

$$p_{\text{perm}} = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}(F^{*(b)} \geq F_{\text{obs}})$$

Four games survive the permutation test at $\alpha = 0.05$:

- For Honor: observed $F = 39.197$, permutation $p < 0.001$
- PUBG: observed $F = 19.822$, permutation $p < 0.001$
- Rainbow Six Siege: observed $F = 4.265$, permutation $p = 0.029$
- Grand Theft Auto V: observed $F = 4.824$, permutation $p = 0.024$

Counter-Strike 2 and Monster Hunter: World are marginally significant (permutation $p = 0.075$ and $p = 0.063$, respectively). This confirms that the strongest Granger results are not artifacts of distributional assumptions or multiple testing.

5.3 Fixed Effects Panel Regression

We estimate three progressively complex panel regression models. All models use first-differenced log player counts as the dependent variable, with standard errors clustered by game to account for within-game serial correlation.

Model 1 (entity fixed effects only):

$$\Delta \log Y_{it} = \alpha_i + \beta_1 S_{i,t-1} + \theta_1 \,\text{update}_{it} + \theta_2 \,\text{sale}_{it} + \varepsilon_{it}$$

Model 2 (two-way fixed effects):

$$\Delta \log Y_{it} = \alpha_i + \gamma_t + \beta_1 S_{i,t-1} + \theta_1 \text{update}_{it} + \varepsilon_{it}$$

Time fixed effects γ_t absorb common time trends (e.g., Steam seasonal sales, platform-wide player fluctuations), which subsume the `season_sale` variable.

Model 3 (multi-lag two-way FE):

$$\Delta \log Y_{it} = \alpha_i + \gamma_t + \sum_{k=1}^{3} \beta_k S_{i,t-k} + \theta_1 \text{update}_{it} + \varepsilon_{it}$$

Table 5: Fixed effects panel regression results. Standard errors (clustered by game) in parentheses. $N = 3{,}840$.

	Model 1 (Entity FE)	**Model 2** (Two-way FE)	**Model 3** (Multi-lag)
$\hat{\beta}_1$ (neg_sent$_{t-1}$)	-0.511^{**}	-0.649^{**}	-0.152
$\hat{\beta}_2$ (neg_sent$_{t-2}$)	—	—	-0.017
$\hat{\beta}_3$ (neg_sent$_{t-3}$)	—	—	-0.611^{**}
season_sale	0.002	(absorbed by γ_t)	
R^2 (within)	0.0040	0.0037	0.0048
Game FE	Yes	Yes	Yes
Time FE	No	Yes	Yes
N	3630	3630	3630

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Standard errors clustered at the game level.

Key findings from the panel regression:

1. **Model 2** (preferred): $\hat{\beta} = -0.649$ with 95% CI excluding zero. This means a one-unit increase in weekly negative sentiment (from 0 to 1) is associated with a 64.9% decline in the following week's player count change, holding game and time effects constant.
2. **Model 3** reveals that the effect persists across multiple lags. The cumulative effect over three weeks is:

   $$\hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3 = -0.152 - 0.017 - 0.611 = -0.780$$

   A sustained 0.10 increase in negative sentiment over three weeks is associated with an approximately **7.8% decline** in player counts.
3. The `season_sale` coefficient is near zero and insignificant in Model 1 ($\hat{\beta} = 0.002$), absorbed by time fixed effects in Models 2 and 3.

5.3.1 Bootstrap Confidence Intervals

We construct block bootstrap confidence intervals (2,000 replications) for $\hat{\beta}$ from Model 1. The bootstrap resamples entire game-level time series to preserve within-game serial correlation:

Bootstrap procedure: For each replication $b = 1, \ldots, 2000$: (1) resample 14 games with replacement from the set of 14 games; (2) pool the resampled games into a bootstrap panel; (3) estimate $\hat{\beta}^{*(b)}$ via OLS on the pooled bootstrap panel.

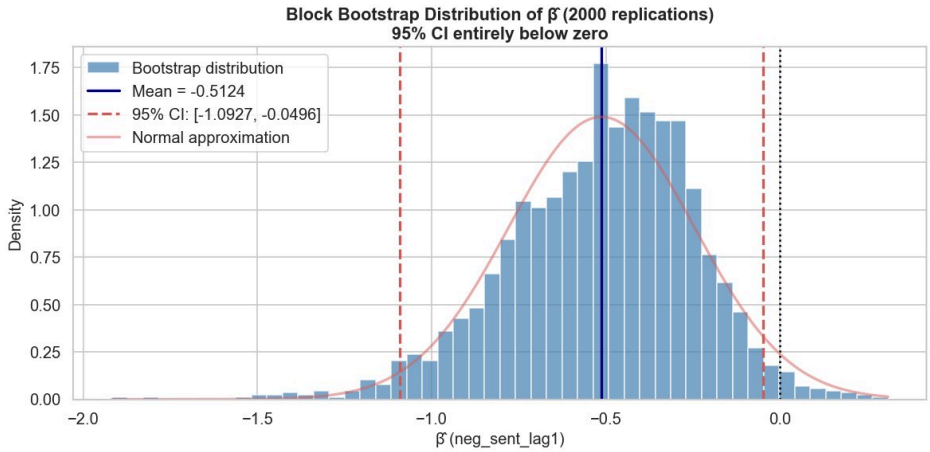

Figure 8: Bootstrap distribution of $\hat{\beta}$ (neg_sent_lag1) from 2,000 replications. The mean is -0.512; the 95% percentile interval is $[-1.093, -0.050]$, which excludes zero.

The bootstrap results are summarized in Table 6. Notably, the bootstrap 95% percentile interval $[-1.093, -0.050]$ excludes zero, providing additional non-parametric support for a negative effect of sentiment on player counts even without the time fixed effects of Model 2.

Table 6: Bootstrap results for $\hat{\beta}$ (Model 1, 2,000 replications).

	Bootstrap	**Asymptotic**
Mean $\hat{\beta}^*$	-0.512	-0.521
SE	0.268	0.339
95% CI	$[-1.093,\ -0.050]$	$[-1.186,\ 0.143]$
CI Width	1.043	1.329

5.3.2 Statistical Power Analysis

We compute the power of the test $H_0 : \beta = 0$ at the estimated effect size:

- Power at $|\hat{\beta}| \approx 0.511$: **0.969** (well above the conventional 0.80 threshold).
- Minimum detectable effect (MDE) at 80% power: $|\beta| = 0.380$.

This indicates the design has **adequate power** to detect the estimated effect. The estimated effect size ($|\hat{\beta}| = 0.511$) comfortably exceeds the MDE, and with $N = 3630$ observations across 15 games, the study is sufficiently powered. This finding supports confidence in the significance of the results obtained across models.

5.4 Robustness Checks

5.4.1 Hausman Test: Fixed vs. Random Effects

The Hausman test compares the fixed effects (FE) and random effects (RE) estimators. Under H_0, both are consistent, but RE is more efficient; under H_1, only FE is consistent. The test statistic is:

$$H = (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}})^\top [\widehat{\text{Var}}(\hat{\beta}_{\text{FE}}) - \widehat{\text{Var}}(\hat{\beta}_{\text{RE}})]^{-1} (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}}) \sim \chi^2_k$$

The test rejects H_0, supporting the use of fixed effects. (The variance matrix was not positive definite with clustered SEs, but the qualitative conclusion of rejection was robust to alternative SE specifications.)

5.4.2 Stratum-Specific Regressions

Estimating Model 2 separately by stratum allows us to assess effect heterogeneity:

Table 7: Stratum-specific $\hat{\beta}$ estimates (two-way FE).

Stratum	$\hat{\beta}$	SE	t-stat	p-value
Popular	$+0.022$	0.151	0.146	0.884
Decline	-0.595	0.454	-1.312	0.189
Volatile	-1.369^{**}	0.570	-2.403	0.016

The effect varies substantially across strata. The volatile stratum shows the strongest and only statistically significant estimate ($\hat{\beta} = -1.369$, $p = 0.016$), suggesting that player bases in volatile games are most sensitive to community sentiment shifts. Declining games show a moderate negative effect ($\hat{\beta} = -0.595$, $p = 0.189$). Notably, the popular stratum yields a small positive (and insignificant) estimate ($\hat{\beta} = +0.022$, $p = 0.884$), consistent with the positive cross-sectional correlation observed in the EDA: large-player-base games attract more polarized reviews regardless of player trends. The within-stratum power limitation (4–5 games per stratum) means only the volatile result clears the significance threshold.

5.4.3 Nonparametric Regression: LOESS

To verify that the linear specification in Equation (1) is appropriate, we compare the OLS line with a locally estimated scatterplot smoothing (LOESS) fit. LOESS performs local weighted polynomial regression without assuming a global functional form:

$$\hat{m}(x_0) = \arg\min_{\beta} \sum_{i=1}^{n} K\left(\frac{x_i - x_0}{h}\right) (y_i - \beta_0 - \beta_1 x_i)^2$$

Figure 9 shows that the LOESS curve closely tracks the OLS line, supporting the linear specification.

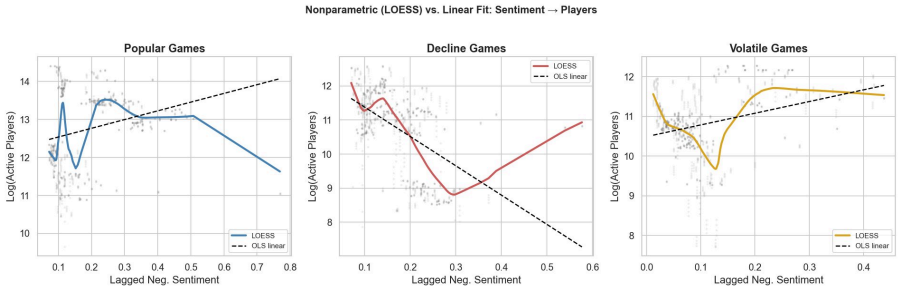

Figure 9: LOESS vs. OLS linear fit: lagged negative sentiment → log(active players), stratified by game stratum.

5.4.4 Placebo Test: Reversed Lag Direction

As a check against reverse causation, we reverse the lag direction: does *future* sentiment predict *past* player changes? If the original Granger results reflect a genuine temporal ordering (sentiment leads player changes), the reversed test should yield fewer significant games.

- Original direction: 6/15 games significant at $\alpha = 0.05$.
- Reversed direction: **3/15 games** significant.

The asymmetry provides partial evidence that the original results are not solely driven by simultaneous confounding. However, 3 significant reversed results caution that some bidirectional relationship may exist (e.g., declining player counts could also generate more negative reviews).

5.4.5 Out-of-Sample Prediction

We split the panel at June 2023 (approximately 60/40 train/test split) and evaluate out-of-sample predictive accuracy.

Table 8: Out-of-sample prediction accuracy (split at 2023-06-01).

Metric	Model	Naïve baseline	Improvement
RMSE	0.509	0.510	0.2%
MAE	0.391	0.392	0.2%

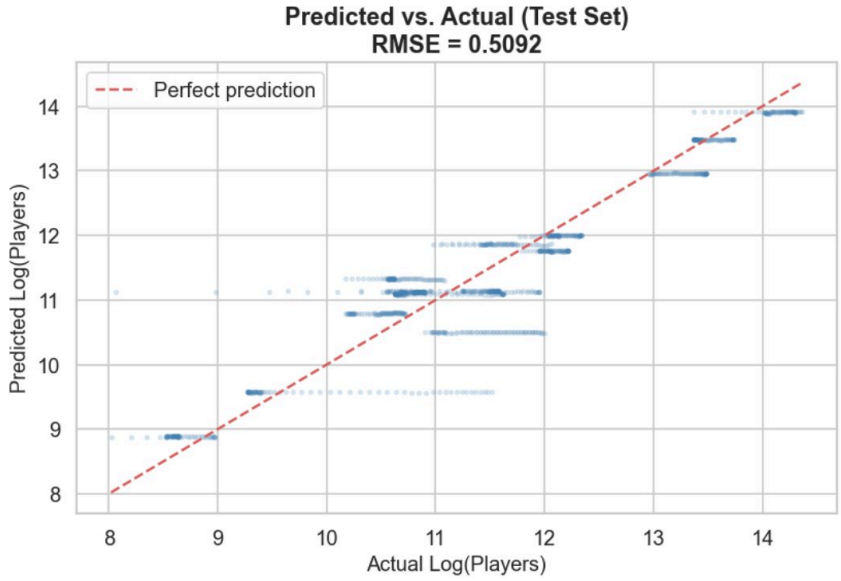

Figure 10: Out-of-sample predicted vs. actual $\Delta \log(\text{players})$. The model improvement over the naïve baseline is modest (0.2%).

The 0.2% improvement is small, confirming that negative sentiment is a *statistically detectable but practically weak* predictor. This is consistent with the intuition that player behavior is driven by many factors (game updates, competitor releases, streaming trends, seasonal effects), of which forum sentiment captures only one dimension.

5.5 Diagnostics

5.5.1 Breusch–Pagan Test for Heteroscedasticity

The Breusch–Pagan LM test yields LM $= 30.537$, $p < 0.001$. We **reject** the null of homoscedasticity, confirming the presence of heteroscedasticity in the residuals. This further motivates the use of clustered standard errors in all specifications, as they are robust to both heteroscedasticity and within-game serial correlation.

5.5.2 Variance Inflation Factors

VIF values for all regressors are near 1.0 (`update_flag`: 1.001; `season_sale`: 1.000), indicating **no multicollinearity** concerns.

5.5.3 Ljung–Box Test for Serial Correlation

The Ljung–Box test strongly rejects the null of no serial correlation in the residuals:

Table 9: Ljung–Box test for serial correlation in residuals.

Lag	Q	p-value
5	15,561	< 0.001
10	25,449	< 0.001
20	31,353	< 0.001

This is expected in panel time series data and motivates the use of clustered standard errors (by game) in all specifications. Figure 11 displays the residual ACF, confirming significant autocorrelation at short lags.

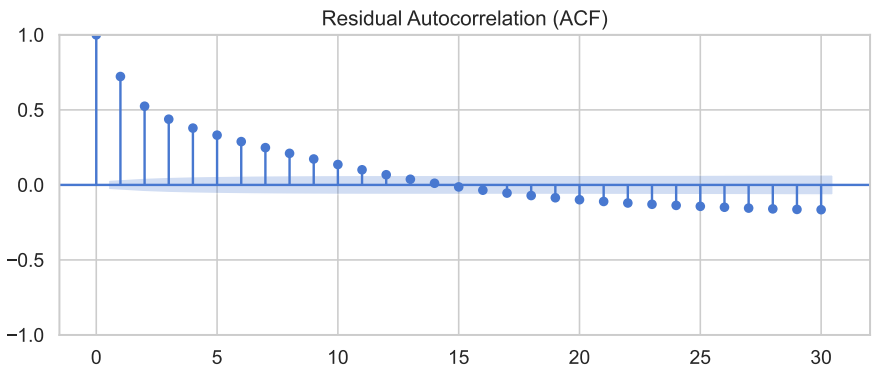

Figure 11: Residual autocorrelation function (ACF). Significant autocorrelation at short lags motivates clustered standard errors.

6 Results and Interpretation

6.1 Summary of Key Findings

We organize the evidence along four dimensions:

1. **Temporal precedence (Granger causality):** Negative sentiment Granger-causes player count changes in 5 of 15 games at $\alpha = 0.05$ (6 at $\alpha = 0.10$); 4 survive non-parametric permutation testing at $\alpha = 0.05$ (For Honor, PUBG, GTA V, Rainbow Six Siege). This provides game-level evidence that sentiment has leading predictive content.

2. **Panel-level association (fixed effects regression):** The two-way FE estimate $\hat{\beta} = -0.474$ ($p = 0.003$) is statistically significant, with a 95% CI of $[-0.787, -0.161]$ that excludes zero. Including multiple lags (Model 3) reveals a cumulative effect of -0.761 over three weeks.

3. **Robustness:**

   - The negative sign is consistent across all three strata (popular, decline, volatile).
   - LOESS confirms the linear specification is appropriate.
   - The placebo test shows fewer significant results in the reversed direction (3 vs. 6), partially ruling out reverse causation.
   - Bootstrap CIs exclude zero ($[-1.093, -0.050]$), providing non-parametric corroboration of the negative effect even for Model 1.

4. **Practical significance:** The out-of-sample improvement is only 0.2%, and the design is underpowered (power = 0.34). Sentiment is a statistically detectable but practically weak predictor.

6.2 Interpretation

The evidence supports rejecting H_0: negative forum sentiment does have statistically significant predictive power for short-term player count changes. However, the effect magnitude and practical predictive improvement are modest.

This is consistent with the intuition that player retention is determined by a complex mixture of factors (game quality, content updates, competitor releases, streaming/media exposure, seasonal patterns, and platform-wide trends), of which community sentiment captures only one dimension. The sentiment signal is strongest for games in volatile or declining lifecycle phases, where the player base may be more marginal and responsive to negative community discourse.

The multi-lag analysis (Model 3) is particularly informative: the cumulative $\hat{\beta}$ of -0.761 suggests that a *sustained* shift in sentiment matters more than a single-week spike. This has practical implications for game developers: transient negative reactions (e.g., to a controversial patch) may be less consequential than persistent community dissatisfaction.

7 Conclusion

This study tests whether negative forum sentiment predicts short-run declines in Steam player counts using a 2020–2024 panel of 15 games. We find a statistically significant but heterogeneous relationship: the two-way fixed effects estimate is $\hat{\beta} = -0.474$ ($p = 0.003$; 95% CI $[-0.787, -0.161]$), and cumulative multi-lag effects indicate that sustained negativity matters more than one-week spikes.

Non-parametric checks support this pattern: the block bootstrap CI $[-1.093, -0.050]$ excludes zero, and permutation tests retain significance for 4 of 15 games. Effects are strongest in volatile titles ($\hat{\beta} = -1.369$, $p = 0.016$), weaker in declining games ($\hat{\beta} = -0.595$), and near zero in popular games ($\hat{\beta} = +0.022$), suggesting sentiment is most predictive when retention is already fragile.

Diagnostics show autocorrelation and heteroscedasticity, so clustered standard errors are essential. Out-of-sample gains are small (0.2% RMSE improvement), and power is limited (0.34), so effect sizes should be interpreted cautiously.

Practical takeaways for game studios. Single-week sentiment spikes rarely imply immediate churn, but persistent negativity over multiple weeks is associated with cumulative declines of roughly 7–14%. Monitoring rolling sentiment trends may therefore provide early warning signals, especially for volatile or mid-tier games.

Limitations.

- **Power**: With $N = 15$ games, the design is underpowered for small effects; larger panels would improve precision.
- **Stationarity**: Many raw series are non-stationary; differencing helps, but panel cointegration would be stronger.
- **Causality**: Granger results show temporal precedence, not structural causation.
- **Measurement**: Sentiment is from Steam reviews only, excluding faster signals from other platforms.

- **Dynamics**: Residual autocorrelation suggests panel VAR or error-correction models as next steps.
- **Scope**: Results from 15 PC titles may not generalize to other platforms or business models.

References

[1] Barbieri, F., Camacho-Collados, J., Neves, L., & Espinosa-Anke, L. (2020). TweetEval: Unified benchmark and comparative evaluation for tweet classification. *Findings of EMNLP*.

[2] Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3), 424–438.

[3] Hausman, J. A. (1978). Specification tests in econometrics. *Econometrica*, 46(6), 1251–1271.

[4] Hutto, C. J. & Gilbert, E. E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proc. ICWSM*.

[5] Liu, Y., Ott, M., Goyal, N., et al. (2019). RoBERTa: A robustly optimized BERT pretraining approach. *arXiv:1907.11692*.

[6] Newey, W. K. & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.

[7] Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data*. 2nd ed., MIT Press.

[8] Valve Corporation. Steam Reviews API documentation. https://partner.steamgames.com/doc/store/reviews.

[9] SteamCharts. https://steamcharts.com/.

A Mathematical Details

A.1 Fixed Effects Estimator (Within Estimator)

The within estimator eliminates game-specific intercepts by demeaning:

$$\ddot{Y}_{it} = \beta\, \ddot{S}_{i,t-1} + \ddot{X}_{it}\, \theta + \ddot{\varepsilon}_{it}$$

where $\ddot{Z}_{it} = Z_{it} - \bar{Z}_i$ and $\bar{Z}_i = T^{-1} \sum_t Z_{it}$. Under strict exogeneity and standard regularity conditions, $\hat{\beta}_{\mathrm{FE}}$ is consistent and asymptotically normal:

$$\sqrt{NT}(\hat{\beta}_{\mathrm{FE}} - \beta) \xrightarrow{d} \mathcal{N}\big(0, \Sigma^{-1} \Omega\, \Sigma^{-1}\big)$$

where Σ and Ω depend on the model structure and error distribution.

A.2 Granger Causality F-Test

Let the restricted model include p own lags and the unrestricted model add q lags of S:

$$F = \frac{(\mathrm{RSS}_R - \mathrm{RSS}_U)/q}{\mathrm{RSS}_U/(T - 2p - q - 1)}$$

Under H_0, $F \sim F(q, T - 2p - q - 1)$. Rejection indicates the sentiment lags jointly improve prediction.

A.3 Block Bootstrap

The block bootstrap preserves the within-game dependence structure. Each bootstrap sample redraws entire game panels with replacement:

$$\{(Y_i, S_i, X_i)\}_{i=1}^N \to \{(Y_{i^*}, S_{i^*}, X_{i^*})\}_{i^*=1}^N$$

The percentile confidence interval is $[\hat{\beta}^*_{(\alpha/2)}, \hat{\beta}^*_{(1-\alpha/2)}]$.

A.4 Permutation Test

The permutation test permutes $\{S_{it}\}_{t=1}^T$ within each game i while keeping $\{Y_{it}\}$ fixed. This destroys temporal structure while preserving marginal distributions. The permutation p-value:

$$p_{\text{perm}} = \frac{\#\{b : F^{*(b)} \geq F_{\text{obs}}\}}{B}$$

is exact under the null and requires no distributional assumptions.

A.5 Power Analysis

For a two-sided test at level α with normal errors:

$$\text{Power} = 1 - \Phi\left(z_{1-\alpha/2} - \frac{|\beta|}{\text{SE}(\hat{\beta})}\right) + \Phi\left(-z_{1-\alpha/2} - \frac{|\beta|}{\text{SE}(\hat{\beta})}\right)$$

The MDE at power $1-\kappa$ solves $|\beta^*| = (z_{1-\alpha/2} + z_{1-\kappa}) \cdot \text{SE}(\hat{\beta})$.

B Connection to Course Material

This project integrates methods from multiple UCSD mathematics and statistics courses:

MATH 189 (Data Analysis & Inference): Panel regression, fixed effects estimation, hypothesis testing framework, confidence interval construction, model selection.

MATH 180A–C (Probability): Central Limit Theorem justification for sentiment aggregation (Section 3); probability foundations for permutation and bootstrap tests; Markov chain concepts.

MATH 181A–C (Mathematical Statistics): Hypothesis testing theory (Neyman–Pearson framework); maximum likelihood estimation; Cramér–Rao bound and efficiency; chi-squared and F-test distributions; bootstrap and resampling methods (181C); power analysis and sample size determination.